Multi-Sensor Coastal Cover Change Analysis Ao Phang Nga Bay, Thailand

Integrating Sentinel-1 SAR and Sentinel-2 Optical Imagery for Multi-Temporal Vegetation Change Detection (2017-2024)

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1. Introduction

Mangrove ecosystems rank among the most productive and carbon-dense coastal habitats on Earth, providing critical services that include coastal protection, fisheries support, and blue carbon sequestration. Despite their ecological importance, these ecosystems face mounting anthropogenic pressure across Southeast Asia. Richards and Friess (2016) documented a regional loss rate of 0.18% per year between 2000 and 2012, driven by aquaculture expansion, rice agriculture, and oil palm plantations, though the relative dominance of each driver varies substantially by country and coastline type.

Thailand presents a particularly instructive case study. After a dramatic decline from 367,900 ha to 160,000 ha between 1961 and 1996, the country achieved notable mangrove recovery, reaching 277,923 ha by 2020 (Chaiklang et al., 2024). This trajectory reflects both the severity of historical degradation and the effectiveness of national conservation and restoration policies. However, recovery is uneven across the coastline, and localised pressure from tourism infrastructure and residual aquaculture continues to affect specific areas, including the Andaman Sea coast.

Ao Phang Nga National Park, located in Phang Nga Province on Thailand's Andaman coast, represents a high-value case: a nationally protected area with extensive mangrove cover under simultaneous pressure from post-tsunami tourism expansion and legacy shrimp aquaculture. Within this park, documented land-use conversion has occurred, with areas historically used for shrimp farming later converted to oil palm and subsequently reclaimed for restoration (IUCN, 2022). The spatial mosaic of loss, stability, and regeneration makes Phang Nga Bay an ideal site for multi-temporal remote sensing analysis.

Optical remote sensing alone is insufficient for robust coastal vegetation monitoring in tropical environments due to persistent cloud cover. This project addresses that limitation by integrating Sentinel-1 C-band Synthetic Aperture Radar (SAR) with Sentinel-2 multispectral imagery, enabling cloud-independent characterisation of vegetation structure alongside spectral vegetation indices. The dual-sensor approach follows a methodological framework established for national-scale mangrove mapping in Thailand (Vongkusolkit et al., 2024), where multi-sensor fusion consistently outperforms single-sensor approaches.

1.1 Objectives

- Quantify vegetation cover change between 2017-2019 and 2022-2024 using spectral indices (NDVI, EVI, NDWI, MNDWI) derived from Sentinel-2 SR composites.
- Characterise structural vegetation change using Sentinel-1 C-band SAR backscatter (VV, VH polarisations) as an independent, cloud-insensitive data source.
- Assess statistical significance of inter-period change across all indices using non-parametric hypothesis testing.
- Demonstrate the added value of multi-sensor fusion over optical-only approaches for coastal vegetation monitoring in a tropical high-cloud environment.

2. Study Area

Ao Phang Nga National Park (8.10-8.75 N, 98.20-98.80 E) covers approximately 400 km2 of coastal and marine habitat in Phang Nga Province, southern Thailand. The bay is characterised by dramatic limestone karst islands, extensive tidal mudflats, seagrass beds, and one of the most intact mangrove systems on the Andaman coast.

The dominant mangrove species include Rhizophora apiculata, Avicennia marina, and Bruguiera gymnorrhiza.

The area experiences a tropical monsoon climate with a pronounced dry season from January to March, which served as the temporal window for Sentinel-2 compositing in this study to minimise cloud contamination. Mean annual rainfall exceeds 2,000 mm, concentrated in the southwest monsoon season (May to November), which would severely compromise optical data quality. This seasonality reinforces the methodological rationale for integrating Sentinel-1 SAR, which is unaffected by cloud cover or precipitation.

Three primary stressors affect vegetation dynamics in the study area: (1) post-tsunami tourism infrastructure expansion since 2004, driven by coastal development along key access corridors; (2) legacy shrimp aquaculture from the 1970s-1990s, which converted substantial mangrove areas to pond systems; and (3) partial mangrove regeneration, supported by national restoration policy and natural recruitment in abandoned aquaculture sites. Poorly planned tourism development has been specifically identified as a driver of mangrove degradation in Southeast Asia through infrastructure clearing and physical trampling of root systems (Mohan et al., 2024). The spatial interaction of these three forces produces the heterogeneous change signal observed in the delta indices.

2017-2024	~400	10 m
7-year span	km2 ROI	Sentinel-1/2
Study Period	Area Analysed	Spatial Resolution

3. Methods

3.1 Data Sources

Two Sentinel satellite data sources were used, both accessed via the Google Earth Engine (GEE) cloud computing platform. Sentinel-2 Surface Reflectance (S2 SR Harmonised, COPERNICUS/S2_SR_HARMONIZED) provided 10 m multispectral imagery for spectral index calculation. Sentinel-1 Ground Range Detected (GRD, COPERNICUS/S1_GRD) provided 10 m C-band SAR backscatter in VV and VH polarisations in Interferometric Wide Swath (IW) mode. Both collections were filtered to two temporal periods: Period 1 (P1: January-March 2017, 2018, 2019) and Period 2 (P2: January-March 2022, 2023, 2024). The January-March window corresponds to the driest months in Phang Nga Province and minimises cloud contamination in the optical composites.

3.2 Sentinel-2 Processing

Images were filtered to a maximum cloud pixel percentage of 10% per scene. Cloud and cloud shadow pixels were masked using the Scene Classification Layer (SCL), removing classes 3 (cloud shadow), 8-10 (cloud medium, high, and cirrus). Reflectance was scaled to surface reflectance in the range 0-1. A pixel-wise median composite was calculated per period from all valid acquisitions. Four spectral indices were computed per period:

- NDVI = (B8 - B4) / (B8 + B4) — the most widely applied index in mangrove remote sensing, used in 82% of peer-reviewed studies (Wan et al., 2022);
- EVI = 2.5 * (B8 - B4) / (B8 + 6*B4 - 7.5*B2 + 1) — reduces atmospheric effects and canopy background noise, used in 28% of mangrove remote sensing studies (Wan et al., 2022);

NDWI = (B3 - B8) / (B3 + B8) — sensitive to moisture content in vegetation canopy;

MNDWI = (B3 - B11) / (B3 + B11) — used as the primary water body mask (threshold MNDWI > 0) applied consistently to both periods.

3.3 Sentinel-1 SAR Processing

Sentinel-1 GRD scenes were filtered to ascending orbit pass for geometric consistency, IW mode, and dual VV+VH polarisation availability. Speckle noise was reduced using a focal mean filter with a 3x3 pixel kernel applied independently to VV and VH bands. A VH/VV ratio band was also calculated as an additional feature sensitive to vegetation structural complexity. Median composites were generated per period over multi-year stacks. The C-band VV polarisation responds to canopy structure and surface scattering in mangrove environments, with typical backscatter values for closed-canopy tropical forest in the range of -7 to -10 dB (Rüetschi et al., 2018; Chularuck et al., 2022).

3.4 Multi-Sensor Stack and Change Detection

For each period, a 13-band multi-sensor stack was assembled by combining 10 Sentinel-2 bands and indices (B2, B3, B4, B8, B11, B12, NDVI, EVI, NDWI, MNDWI) with 3 Sentinel-1 bands (VV, VH, VH/VV ratio). Water pixels were excluded using a union MNDWI mask. NDVI change (delta NDVI = P2 - P1) and SAR backscatter change (delta VV = P2 - P1) were computed and classified into: vegetation gain (delta NDVI > +0.10), stable (-0.10 to +0.10), and vegetation loss (delta NDVI < -0.10). EVI outliers outside the physically valid range [-1, 1] representing 0.038% of P1 and 0.013% of P2 pixels were clipped prior to statistical analysis.

3.5 Statistical Analysis

Inter-period differences in NDVI, EVI, and SAR VV backscatter were assessed using the Wilcoxon rank-sum test (Mann-Whitney U), a non-parametric test appropriate for the non-normal distributions typical of satellite-derived raster data. Random samples of 50,000 pixels per index per period were drawn (seed = 42). Statistical significance was set at alpha = 0.05. Area estimates for each change category were calculated from pixel counts at 10 m resolution (0.01 ha per pixel). All analyses were conducted in R using the terra, sf, trend, and ggplot2 packages.

4. Results

4.1 NDVI Distribution by Period

The NDVI distribution across vegetation pixels shows high mean values in both periods, consistent with dense tropical vegetation cover. Mean NDVI was 0.818 in P1 (2017-2019) and 0.817 in P2 (2022-2024), with median values of 0.868 and 0.867 respectively. These values are consistent with the dense mangrove category (NDVI 0.73-0.81) reported in Sentinel-2 studies (Hossain et al., 2024), and with dense closed-canopy tropical forest. The near-identical means mask a subtle distributional shift: the P2 density curve shows a marginally heavier left tail (lower NDVI values), indicating that a subset of pixels experienced moderate vegetative decline not captured by the mean alone.

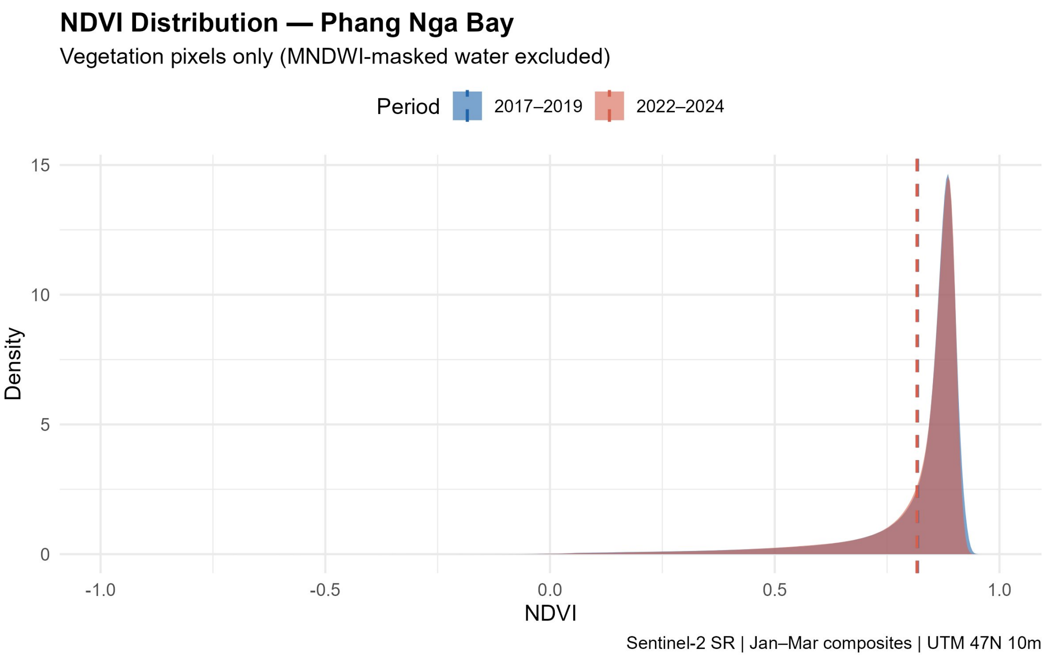

Figure 1. NDVI density distributions for vegetation pixels (MNDWI-masked water excluded) in Period 1 (2017-2019, blue) and Period 2 (2022-2024, red). Dashed lines indicate period means. Sentinel-2 SR, January-March composites, UTM Zone 47N, 10 m resolution.

4.2 Spatial Pattern of NDVI Change

The delta NDVI map reveals a spatially heterogeneous pattern of change across the bay. The western coastal fringe shows the strongest concentration of vegetation loss pixels (delta NDVI < -0.10), consistent with areas of documented legacy aquaculture and coastal development pressure. Vegetation gain is dispersed across the northern sector, suggesting natural regeneration in previously disturbed areas. The central bay is appropriately masked as water. The dominance of stable (yellowish) tones confirms that 84.7% of vegetated land area did not undergo detectable change during the study period.

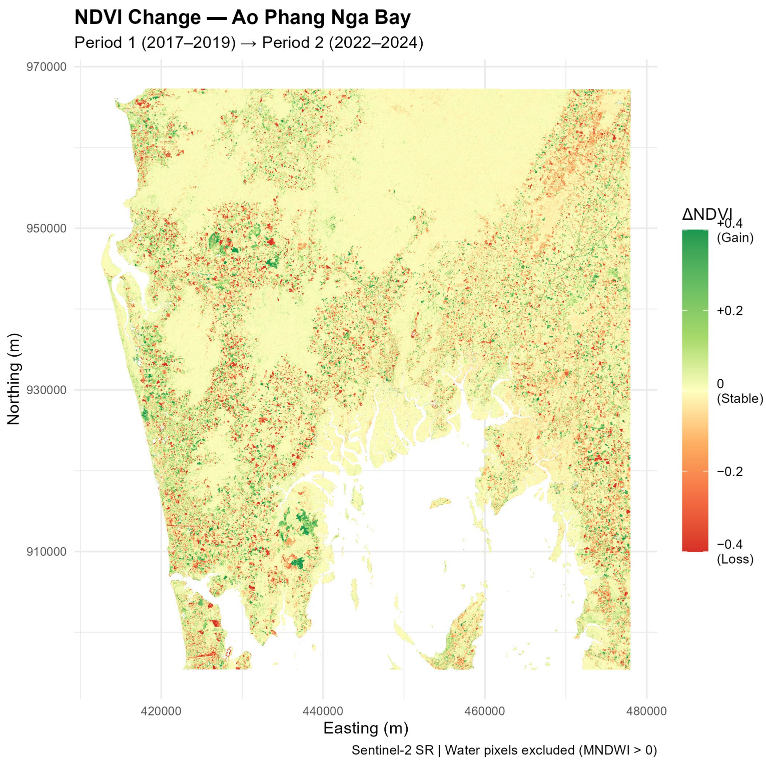

Figure 2. Spatial distribution of NDVI change (delta NDVI = P2 - P1) across Ao Phang Nga Bay. Red tones indicate vegetation loss; green tones indicate gain; yellow indicates stable cover. Water pixels excluded using MNDWI threshold. Sentinel-2 SR, 10 m resolution.

4.3 SAR Backscatter Change

The delta SAR VV map provides a structurally independent characterisation of vegetation change. Mean VV backscatter was -8.31 dB in P1 and -8.37 dB in P2, consistent with closed-canopy tropical forest values reported by Rüetschi et al. (2018) for deciduous forest (-7.75 to -8 dB) and by Chularuck et al. (2022) for mangrove stages in Thailand. The SAR change pattern is spatially more diffuse than the NDVI delta, reflecting the distinct physical mechanism of C-band backscatter: VV responds to canopy structure, stem density, and dielectric properties including moisture content, while NDVI captures photosynthetically active radiation absorption at the canopy surface. Partial spatial divergence between the two change maps is therefore expected and constitutes a methodological strength: it demonstrates that optical and SAR sensors capture complementary dimensions of vegetation dynamics.

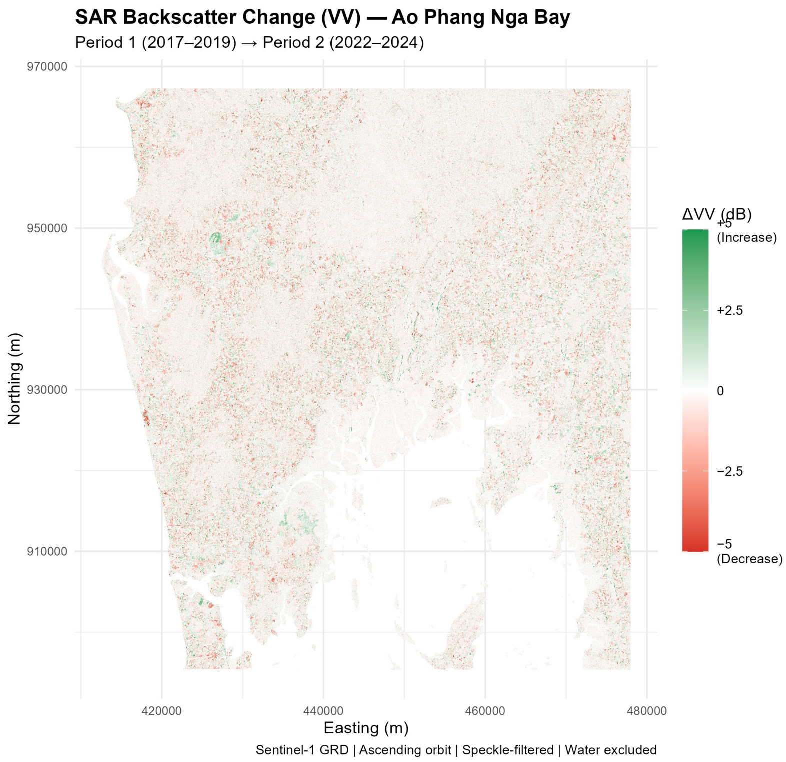

Figure 3. Spatial distribution of SAR VV backscatter change (delta VV = P2 - P1) across Ao Phang Nga Bay. Red tones indicate backscatter decrease (structural loss); green tones indicate increase (structural gain). Sentinel-1 GRD, ascending orbit, speckle-filtered, water pixels excluded. 10 m resolution.

4.4 Area-Based Change Quantification

Based on the NDVI differencing threshold (|delta NDVI| > 0.10), the study area exhibited a nearly symmetric pattern of gain and loss. Vegetation loss affected 26,918 ha (7.8% of vegetated area), vegetation gain was detected across 25,922 ha (7.5%), and 292,234 ha (84.7%) showed no meaningful change. Net vegetation loss was approximately 996 ha. This near-symmetry is ecologically meaningful: it suggests a dynamic mosaic in which degradation at one location is partially offset by natural regeneration elsewhere, consistent with community-based restoration efforts documented nationally by Chaiklang et al. (2024).

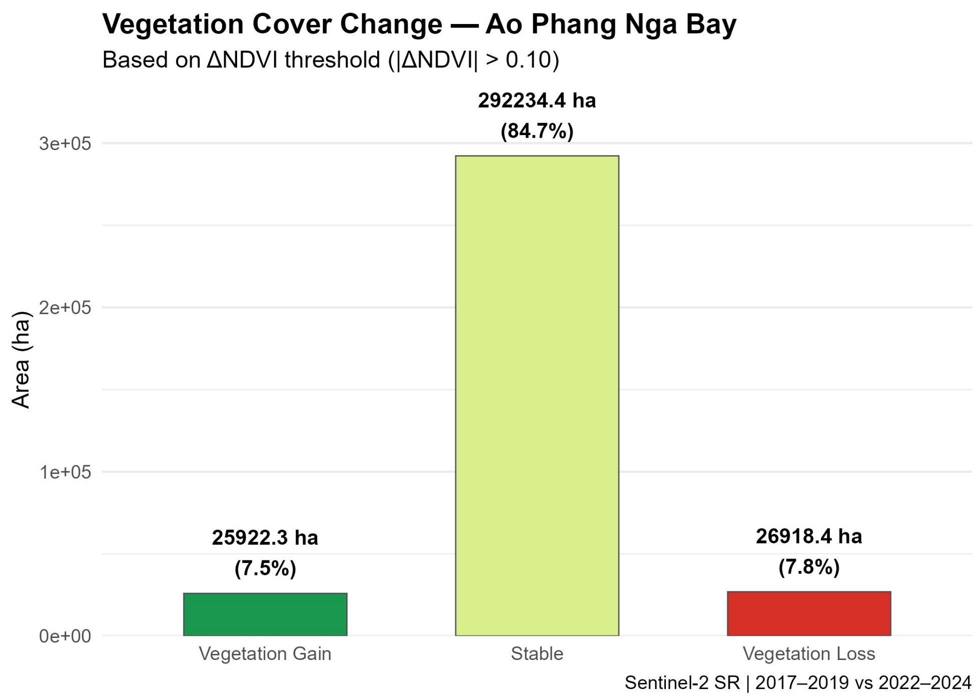

Figure 4. Area (ha) and percentage of vegetated land classified as Vegetation Gain, Stable, and Vegetation Loss based on NDVI differencing threshold |delta NDVI| > 0.10. Sentinel-2 SR composites, 2017-2019 vs. 2022-2024.

Category	Area (ha)	% Vegetated Area	NDVI Threshold
Vegetation Gain	25,922	7.5%	delta NDVI > +0.10
Stable	292,234	84.7%	-0.10 to +0.10
Vegetation Loss	26,918	7.8%	delta NDVI < -0.10
Net Change	-996	-0.3%	Gain minus Loss

4.5 Multi-Index Comparison and Statistical Testing

Violin and boxplot distributions of NDVI, EVI, and SAR VV confirm consistent inter-period patterns across all three independent indicators. EVI distributions show greater spread in P2, indicating increased spatial heterogeneity in vegetation vigour, a pattern associated with landscape fragmentation rather than uniform decline. SAR VV distributions are broadly stable between periods with slightly heavier lower tails in P2, consistent with localised structural change in canopy density.

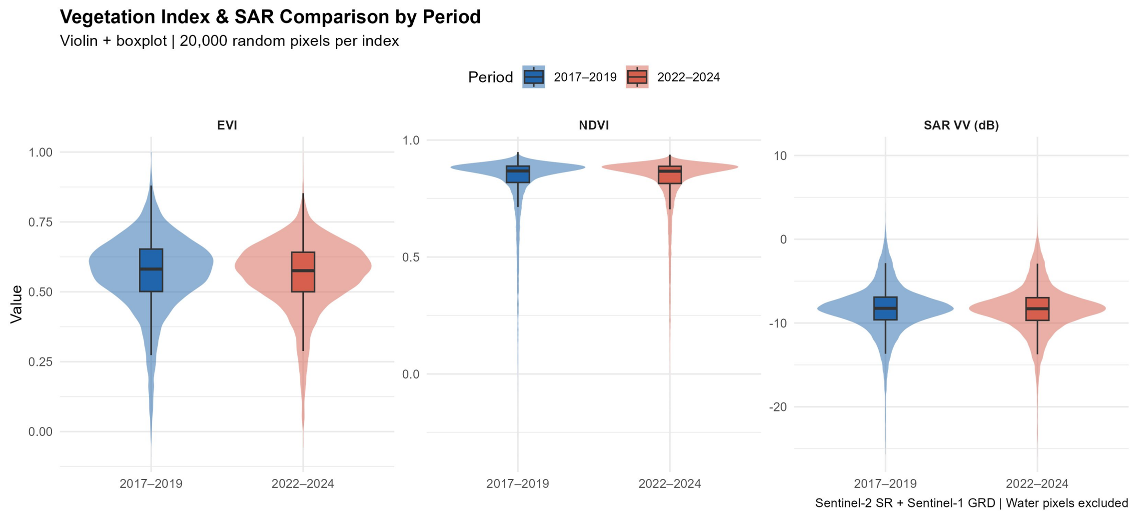

Figure 5. Violin and boxplot distributions of EVI, NDVI, and SAR VV backscatter for Period 1 (blue) and Period 2 (red). Distributions based on 20,000 random pixels per index. Sentinel-2 SR + Sentinel-1 GRD. Water pixels excluded.

Wilcoxon rank-sum tests confirmed that inter-period differences are statistically significant for all three indices (Table 2). The smallest effect was observed in NDVI (p = 5.70 x 10-11), consistent with the near-identical means and the known saturation of NDVI in dense tropical canopy conditions. EVI showed the most highly significant result (p = 5.82 x 10-34), reflecting its greater sensitivity to canopy structural variation in high-biomass forests. SAR VV also returned a significant result (p = 2.44 x 10-5), independent of the optical signal, providing cross-sensor validation that the detected change is real and not an artefact of composite composition or atmospheric correction.

Index	W Statistic	p-value	Significant (a=0.05)
NDVI	1,279,902,552	5.70 x 10-11	Yes
EVI	1,305,451,763	5.82 x 10-34	Yes
SAR VV	1,269,262,675	2.44 x 10-5	Yes

Index	Mean P1	Mean P2	Delta	SD P1	SD P2
NDVI	0.818	0.817	-0.001	0.141	0.137
EVI	0.567	0.558	-0.009	0.137	0.127
NDWI	-0.720	-0.728	-0.008	0.107	0.105
SAR VV (dB)	-8.31	-8.37	-0.060	3.00	2.96

Table 2. Wilcoxon rank-sum test results for NDVI, EVI, and SAR VV between periods (n = 50,000 random pixels per index). Table 3. Summary statistics for all indices by period. EVI clipped to physical range [-1, 1] prior to analysis.

Key Finding: All three independent indicators — NDVI (optical), EVI (optical), and SAR VV (microwave) — detected statistically significant inter-period change (p < 0.05), confirming that the observed vegetation dynamics are real and not artefactual. The convergence of optical and SAR signals constitutes cross-sensor validation of the change detection results.

5. Discussion

5.1 Ecological Interpretation

The near-symmetric pattern of vegetation gain (25,922 ha, 7.5%) and loss (26,918 ha, 7.8%) with a net loss of approximately 996 ha reflects a dynamic coastal mosaic rather than a unidirectional deforestation trend. This finding aligns with national-scale observations by Chaiklang et al. (2024), who documented that Thailand's mangrove area expanded from 160,000 ha to 277,923 ha between 1996 and 2020, demonstrating that spatially concurrent loss and gain is the expected pattern for a partially recovering system under heterogeneous pressure.

The concentration of loss pixels along the western coastal fringe is spatially consistent with the documented history of shrimp aquaculture conversion in Phang Nga Province and with coastal tourism infrastructure expansion following the 2004 Indian Ocean tsunami. Poorly planned tourism development has been identified as a driver of mangrove loss in Southeast Asia through infrastructure clearing, boat traffic, and physical trampling of pneumatophore root systems (Mohan et al., 2024). The dispersed gain signal in the northern sector is consistent with natural regeneration dynamics in transitional zones where land-use pressure has reduced, and with national restoration efforts in abandoned aquaculture sites.

The mean NDVI of 0.817-0.818 across both periods is consistent with values reported for dense mangrove and closed-canopy tropical forest. Dense mangrove categories in Sentinel-2 studies report NDVI in the range 0.73-0.81 (Hossain et al., 2024), and healthy mangrove stands in Indonesia showed NDVI exceeding 0.9 prior to documented decline. That the dominant vegetation cover maintains values in the 0.81-0.87 median range across both periods supports the interpretation that large-scale structural integrity is maintained, with detected changes concentrated in marginal and transitional zones.

5.2 Multi-Sensor Validation

The partial spatial divergence between the NDVI delta and SAR VV delta maps is a methodologically important result. While both sensors detect statistically significant inter-period change, they do not delineate identical spatial patterns. This is expected given the distinct physical mechanisms: NDVI captures photosynthetically active radiation absorption at the canopy surface, while C-band VV backscatter responds to canopy structure, stem and branch density, and dielectric properties modulated by moisture content (Chularuck et al., 2022). The convergence of all three indices on statistically significant change provides multi-source evidence that the detected dynamics are ecologically real, consistent with the ensemble approach of Vongkusolkit et al. (2024), who demonstrated that multi-sensor fusion achieves substantially higher accuracy for mangrove mapping in Thailand than single-sensor methods.

5.3 Limitations

Three methodological limitations should be acknowledged. First, residual cloud contamination was visually detectable in both Sentinel-2 composites despite the 10% cloud filter and January-March seasonal restriction. Residual cloud shadow pixels can introduce spurious low-NDVI values contributing to the observed loss signal; SAR data, being cloud-independent, is not subject to this limitation. Second, EVI outliers representing 0.038% (P1) and 0.013% (P2) of pixels were identified outside the physical range [-1, 1], attributable to residual atmospheric artefacts; these were clipped prior to analysis. Third, the absence of ground truth validation data prevents formal accuracy assessment of the change classification, which should therefore be considered exploratory.

Key Finding: The partial divergence between optical NDVI and SAR VV change patterns is a finding, not a limitation: optical indices detect canopy greenness while SAR detects structural and dielectric change. Together, they provide a more complete characterisation of coastal vegetation dynamics than either sensor alone.

6. Conclusions

This study integrated Sentinel-1 SAR and Sentinel-2 multispectral imagery to characterise five-year coastal vegetation dynamics in Ao Phang Nga Bay, Thailand. The multi-sensor approach enabled cloud-robust change detection in a tropical environment where optical data alone is insufficient for consistent time-series analysis.

Key findings: (1) 84.7% of vegetated area remained stable between 2017-2019 and 2022-2024; (2) net vegetation loss was modest at approximately 996 ha, reflecting a dynamic mosaic of localised loss and concurrent regeneration; (3) all three independent indices (NDVI, EVI, SAR VV) confirmed statistically significant inter-period change, providing cross-sensor validation; (4) EVI showed the strongest statistical signal ($p = 5.82 \times 10\text{-}34$), supporting its use as a complementary index to NDVI in high-biomass tropical environments.

The methodology demonstrated here — dual-period median compositing, MNDWI-based water masking, 13-band multi-sensor stacking, and cross-sensor statistical testing — is transferable to other tropical coastal systems. Future work should incorporate Random Forest classification of the multi-sensor stack to produce discrete land-cover change maps with formal accuracy assessment using ground truth data.

~996	84.7	3 / 3
ha (2017-2024)	% of vegetated area	$p < 0.05$
Net Vegetation Loss	Area Stable	Indices Significant

7. References

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